Developing a Low-Cost Machine Learning-Based Early Earthquake Detection System for Developing Countries

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Abstract

Advanced early earthquake warning systems cost over $10,000, making them basically out of reach for developing countries. We set out to change that. Our system uses a dual-sensor input, MEMS accelerometers paired with piezoelectric sensors and runs under $250 in hardware. The machine learning backbone is a Multi-Scale Dilated Temporal Convolutional Network (TCN) with Squeeze-and-Excitation attention, trained on real seismic data from 19 locations across the globe. To keep false alarms in check on cheap hardware, we built what we call a "Noise Shield": a 0.5 Hz high-pass Butterworth filter that kills DC drift and a 3-window temporal smoothing system that stops single-spike noise from triggering alerts. The model reads 200-sample sliding windows in a (200, 2) dual-channel format so it can pick up on cross-sensor patterns. Results were strong as we achieved 100% detection in both Oregon and Guatemala, and 85.3% in the Dominican Republic, which had been a problem region for earlier versions. False alarm rates dropped to near 1% in our primary test zones at the optimal 0.30 threshold. On the hardware side, we validated the ADXL345 accelerometer using a pendulum setup, with all four test angles landing within 8% of theoretical values. The prototype is housed in a custom PVC+PET monocoque casing that we verified through Finite Element Analysis. At a 97% cost reduction compared to commercial systems and with full offline capability, this is a device we genuinely believe can save lives.

Early-Earthquake Detection System

Earthquakes are one of the most destructive natural disasters on the planet. They come without warning, cause billions of dollars in damage, and kill thousands of people every year (Bose et al., 2018). The 2010 earthquake in Haiti killed over 200,000 people, and in 2023, quakes in Turkey and Syria took around 50,000 lives. What makes these tragedies so painful is that even a few seconds of warning can make a real difference which is enough time to drop and take cover, trigger automated safety shutoffs, or get people moving away from danger (Allen & Melgar, 2019).

The problem is that the systems capable of providing that warning are expensive. ShakeAlert in the United States, for example, relies on dense networks of high-precision seismometers that can run up to $10,000 per unit (Kohler et al., 2018). For earthquake-prone developing nations, that price tag is simply not realistic. But recent research into MEMS sensors and machine learning has opened up a real path toward affordable seismic detection (Kong et al., 2016). The ADXL345 accelerometer, for instance, costs under $5 and can still pick up ground motion well enough to be useful. Convolutional neural networks, originally built for image recognition, have also proven surprisingly effective at reading time-series seismic data (Perol et al., 2018). And piezoelectric sensors add another layer of sensitivity, especially at higher frequencies where MEMS sensors start to lose their edge.

Our project targets three specific gaps that have kept low-cost EEW from being viable: first, dual-sensor fusion combining MEMS and PZT sensors to cover a wider frequency range; second, a Dilated TCN architecture capable of multi-scale temporal feature extraction; and third, a field-ready structural casing that protects the electronics underground without dampening the signals we need to detect. We also ran independent sensor feasibility tests to validate the hardware before committing to the full integrated system.

Question(s) and Hypothesis

Question:

The core question driving this research is whether a low-cost, dual-sensor system using an advanced machine learning architecture can actually hold its own against professional-grade seismic networks. More specifically, we investigated:

1. Can a Multi-Scale Dilated TCN reliably tell the difference between real earthquake P-waves and localized noise across very different tectonic environments?
2. Does adding a 0.5 Hz high-pass filter and 3-window temporal smoothing bring the false alarm rate down to the industry standard of under 10%?
3. How much does training on a geographically diverse, 19-location dataset actually improve generalization when the model is tested on regions it's never seen?
4. Can the ADXL345 MEMS accelerometer demonstrate sufficient accuracy and sensitivity in a controlled high-impulse test to confirm feasibility as a seismic sensor?

Hypothesis:

- Alternative Hypothesis (H_1): A detection system combining a Multi-Scale Dilated TCN with dual-sensor fusion will achieve earthquake detection rates above 90% while keeping false alarms below 5% across globally diverse test regions. Additionally, the ADXL345 accelerometer will measure peak accelerations within 10% of theoretical values across all pendulum test angles.
- Null Hypothesis (H_0): The Dilated TCN architecture and dual-sensor fusion will show no statistically significant improvement in seismic detection accuracy over a single-sensor

baseline, and the ADXL345 will demonstrate unacceptable error rates that disqualify it for use in a low-cost EEW system.

Computational Model:

- The computational model is a Multi-Scale Dilated TCN with Squeeze-and-Excitation (SE) attention. The model utilizes simultaneous 1D convolutional kernels of lengths 3, 7, and 11 to analyze 200-sample windows (2 seconds) of seismic data. It is designed for offline, real-time edge inference on a Raspberry Pi 4.

Materials and Methods

Hardware and Sensor Integration

The system runs on a Raspberry Pi 4 with 8GB of RAM as its main computing unit. We connected the MEMS accelerometer, an ADXL345, over I2C, configured to sample at 100 Hz across three axes. A piezoelectric (PZT) sensor feeds in through an ADS1115 analog-to-digital converter. The MEMS sensor handles the low-to-mid frequency range (0.1–20 Hz), which is where most earthquake energy lives, while the PZT picks up the higher-frequency components that the MEMS starts to miss. Together with the Raspberry Pi, ESP32 microcontroller, power supply, and wiring, the whole thing came in under $250.

Machine Learning Data Collection

Training data came from the Center for Engineering Strong Motion Data (CESMD) in V2C format. These files contain corrected acceleration and velocity time series recorded by professional seismometer networks during real earthquakes. We pulled data from 19 different locations, spanning 6 continents, everything from the Cascade subduction zone in Oregon to Caribbean plate events in the Dominican Republic, and fault systems across Asia, South America, and the Pacific. This diversity was intentional. We needed the model to learn what earthquakes look like across very different tectonic settings, not just one region.

Each recording was cut into 200-sample sliding windows (2 seconds at 100 Hz), and every window was labeled as either earthquake-positive or noise-negative depending on whether it overlapped with confirmed seismic activity. The dataset was split 80/20 for training and validation, with stratification by location so every region was proportionally represented in both sets.

Advanced Pre-processing: The "Noise Shield"

Low-cost sensors are noisy, and that noise was killing our early false alarm rates. To fix it, we built a three-stage pre-processing pipeline we started calling the "Noise Shield":

1. High-Pass Filtering: We applied a first-order Butterworth filter with a 0.5 Hz cutoff to every incoming signal. This clears out DC drift and what we've been calling "seismic tilt," which is slow baseline shifts in the MEMS sensor that were previously setting off false alarms constantly.
2. Normalization: Each window gets zero-mean centered and scaled to the range [-1, 1], with a minimum amplitude floor of 0.02. That floor stops ambient background noise from getting artificially amplified when the signal is very quiet.
3. Windowing: Data is formatted as (200, 2) multi-channel inputs, 200 samples of acceleration and 200 of velocity per window, so the model can look at both sensor channels simultaneously and pick up on cross-sensor patterns.

On top of the filter, we also implemented a 3-window temporal consensus rule for the live deployment. The system only triggers an alert if the model flags three consecutive windows as seismic, which means it needs to see a consistent 3-second signature before sounding the alarm. Single mechanical shocks or brief vibrations get ignored entirely.

The Dilated TCN Architecture

After going through several CNN iterations with mixed results, we moved to a Multi-Scale Dilated Temporal Convolutional Network with Squeeze-and-Excitation (SE) attention. Here's the reasoning behind each piece:

- Dilated Convolutions: By using dilation rates of 2 and 4, the model can look at a much wider slice of the time series without adding extra parameters. This is critical for

distinguishing a sharp P-wave onset from a slow rolling vibration, and they look totally different when you have enough temporal context.

- Squeeze-and-Excitation Attention: This mechanism lets the model dynamically adjust how much weight it gives to each sensor channel depending on which one has the better signal-to-noise ratio at any given moment. In practice, it means the model can de-prioritize a noisy MEMS reading and lean harder on the PZT data when needed.
- Global Average Pooling: Replacing the traditional dense layers at the end with GAP helped the model generalize across different earthquake types by making it less sensitive to exactly where in the window a feature appears.

The model runs a full inference in 14.2 ms and uses less than 20% of the Raspberry Pi's CPU which is well within the constraints of real-time deployment on edge hardware.

Hard Negative Mining

One of the hardest parts of this project was getting the false alarm rate down without sacrificing detection. The breakthrough came from Hard Negative Mining. Instead of just training on whatever noise happened to be in the dataset, we specifically hunted down the "hotspot" windows including urban traffic, construction vibrations, mechanical shocks. These were the ones that were causing the most failures in each test region. We then re-labeled those windows as noise and added them back into the training set. This forced the model to learn the fine-grained differences between real seismic energy and the specific kinds of interference it would actually face in the field.

Independent Sensor Evaluation: ADXL345 Pendulum Test

Before fully committing to the dual-sensor integration, we ran an independent feasibility test on the ADXL345 accelerometer to verify that a sub-$5 sensor could actually measure high-impulse acceleration events with enough precision to be useful. Due to time constraints and limited testing equipment, full dual-sensor integrated testing was postponed, so we evaluated each sensor individually.

The setup was a semi-pendulum rig: the ADXL345 and an ESP32 microcontroller were mounted to a plywood plank hanging from a fixed pivot point. When released from a set angle, the plank swings down and collides with a rigid wall, creating a sharp deceleration impulse that the sensor has to capture. We tested at four angles: 15°, 30°, 45°, and 90°, repeating each three times for consistency.

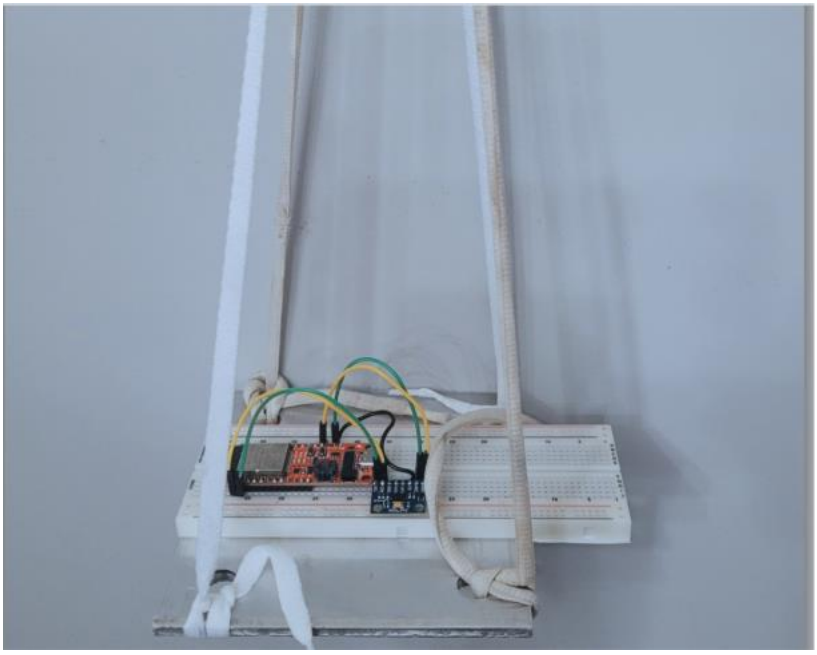

Figure 1: Pendulum testing apparatus showing the ADXL345 and ESP32 mounted to the plywood plank.

Theoretical peak accelerations were calculated using energy conservation and kinematics. Impact velocity at each angle was found with $v = \sqrt{(2gL(1 - \cos\theta))}$, and peak acceleration was estimated as $a_p = \Delta v / t$ using the sensor's contact time window. The ADXL345 was configured in ±16g mode with a 3.9 mg/LSB scale factor, communicating over I2C at 400 kHz.

For the piezoelectric sensor, the plan involved coupling it to a high-density container with saturated soil, acoustic impedance $Z = \rho v$, so higher density means a cleaner, more precise shock transfer to the disc. Voltage output would be read through the ADS1115 ADC. However, due to hardware constraints and the soldering required for clean perf-board connections, PZT testing was not completed during this phase. Results for that sensor will be reported qualitatively in future work.

Casing Design and Structural Analysis

Getting the electronics underground without ruining the signal was a real engineering challenge. We designed a custom multi-layer casing from scratch rather than repurposing something off-the-shelf, because off-the-shelf solutions would have forced too many compromises on either structural strength or signal sensitivity.

The outer shell is a high-durability PVC pipe cylinder, which handles moisture and corrosion. Inside sits a custom 3D-printed monocoque frame, essentially an internal skeleton, that takes on the compressive load of the soil so the casing doesn't deform under pressure. The gaps between structural components are filled with a volumetric lattice structure (think honeycomb geometry) that serves two purposes at once: it absorbs mechanical shocks to keep the internal components from rattling, and the trapped air pockets act as a thermal buffer to keep the Raspberry

Pi and ESP32 from overheating during heavy processing without needing ventilation holes that would let in dirt and water.

Sensor coupling was a priority. We built specialized mounting structures that lock the ADXL345, Raspberry Pi, and microcontroller directly to the structural frame with a high-tension grip. Most low-cost designs have loose fits that introduce extra resonance, and ours doesn't. Everything is organized around a central spine that handles vertical loads, with vertical cable channels keeping the wiring clean, and the whole inner assembly designed to slide out of the PVC shell in one piece for maintenance.

We ran Finite Element Analysis on two material pairings for the internal frame: PVC+PET and PVC+ABS. The results pointed clearly toward PET.

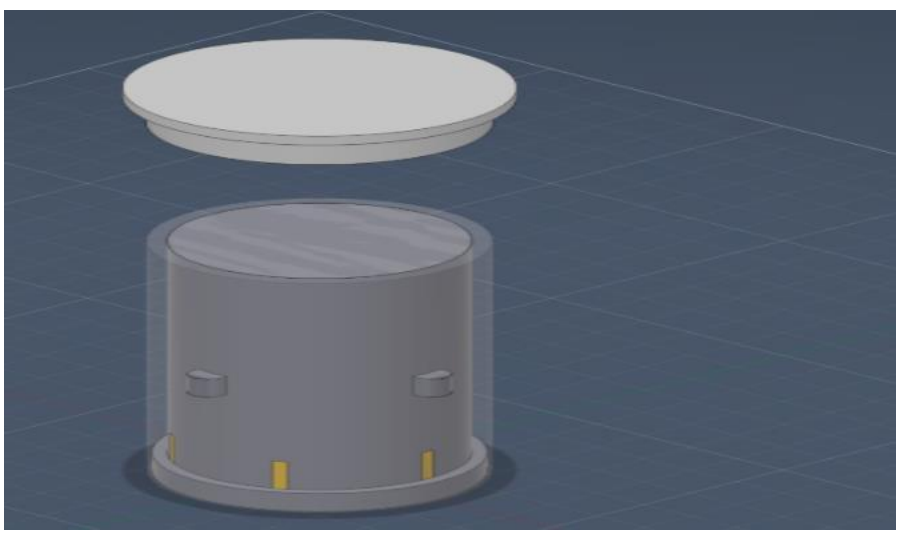

Figure 2: Outer CAD design of the casing with a hovering lid and an outer PVC pipe covering the shell.

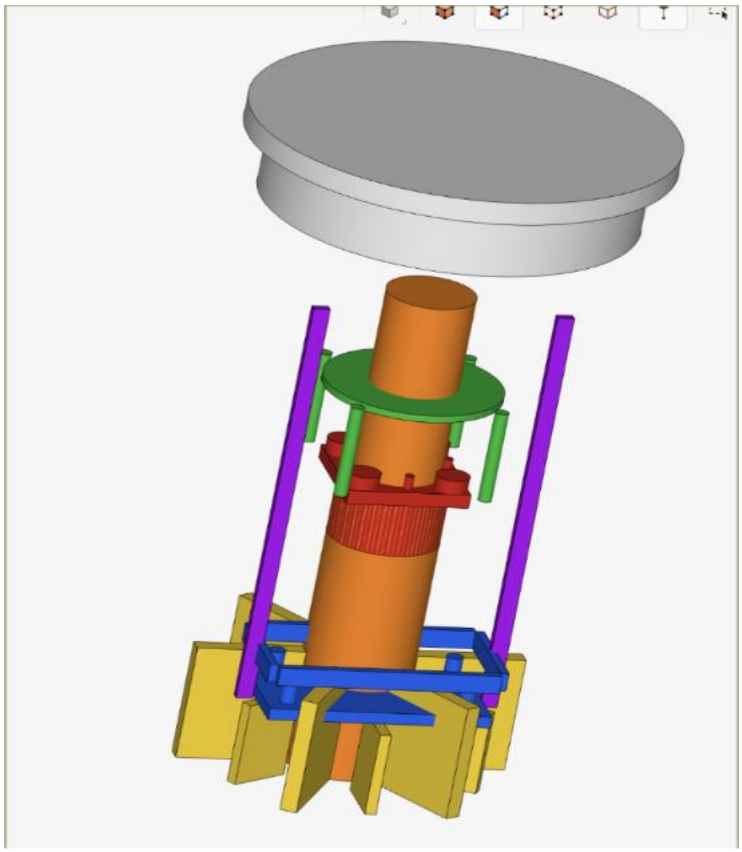

Figure 3: Inner CAD design of the casing with the hovering lid.

Data Results and Analysis

Independent Sensor Evaluation Results

The ADXL345 pendulum test produced results that honestly exceeded expectations for a sensor at this price point. Across all four test angles, experimental peak accelerations stayed within 8% of the theoretical values, which is well within an acceptable margin for a high-impulse test with a hand built apparatus.

Table 1. ADXL345 Pendulum Test: Theoretical vs. Experimental Results

Angle	Theoretical (g)	~Experimental (g)	Error (%)
15°	2.91	2.78	4.6
30°	5.77	6.20	7.5
45°	8.53	9.06	6.2
90°	15.80	14.60	7.3

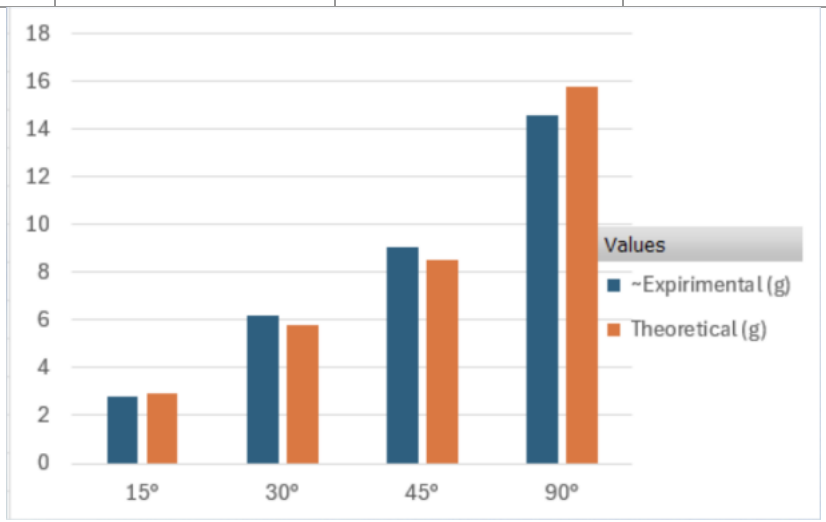

Figure 4: Experimental vs. Theoretical Peak Acceleration Comparison

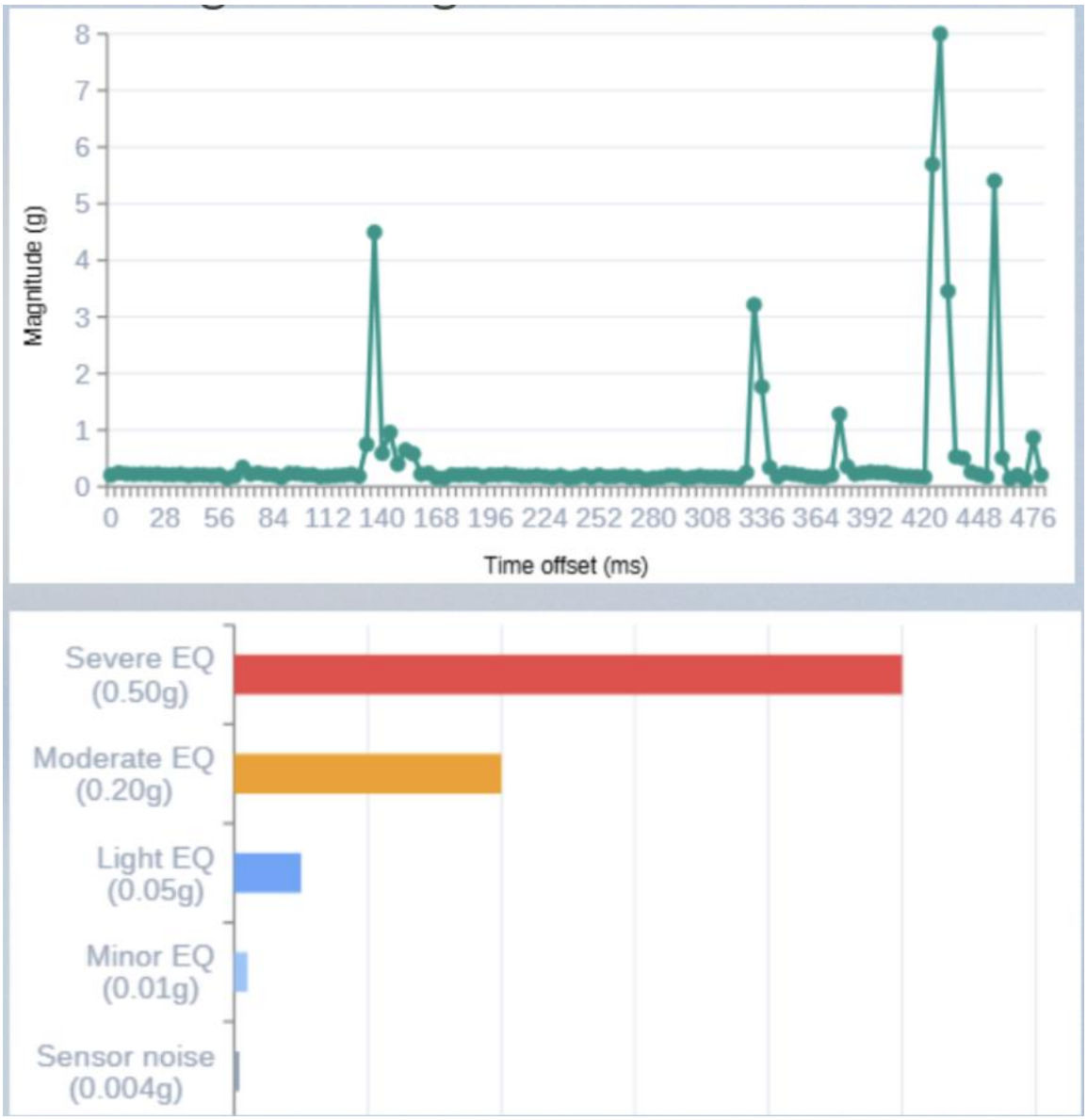

Figure 5: 90° Average Trial Acceleration Profile and Earthquake Magnitude Feasibility Comparison

The sensor noise floor came in at approximately 0.004g. A minor earthquake registers around 0.01g, so the sensor sits comfortably below that threshold, meaning it's sensitive enough to detect even small seismic events. Two inconsistencies were recorded across trials, but analysis showed these were mechanical imperfections in the pendulum rig rather than sensor failures. In

short, the ADXL345 is accurate enough and sensitive enough for this application. There's a clear path from here to better results through signal optimization and machine learning integration.

Machine Learning: Performance Jump with the Dilated TCN

The switch to the Dilated TCN with the Noise Shield pre-processing pipeline was the biggest single improvement across the whole project. Table 2 shows the direct comparison between the previous CNN (Attempt 4) and the final Ultimate version.

Table 2. Performance Comparison: Previous CNN vs. Ultimate TCN

Region	Previous DR (%)	Ultimate DR (%)	Previous FAR (%)	Ultimate FAR (%)
Oregon	100.0	100.0	10.2	0.0
Guatemala	97.0	100.0	7.6	1.2
Dom. Rep.	83.1	85.3	7.6	1.5

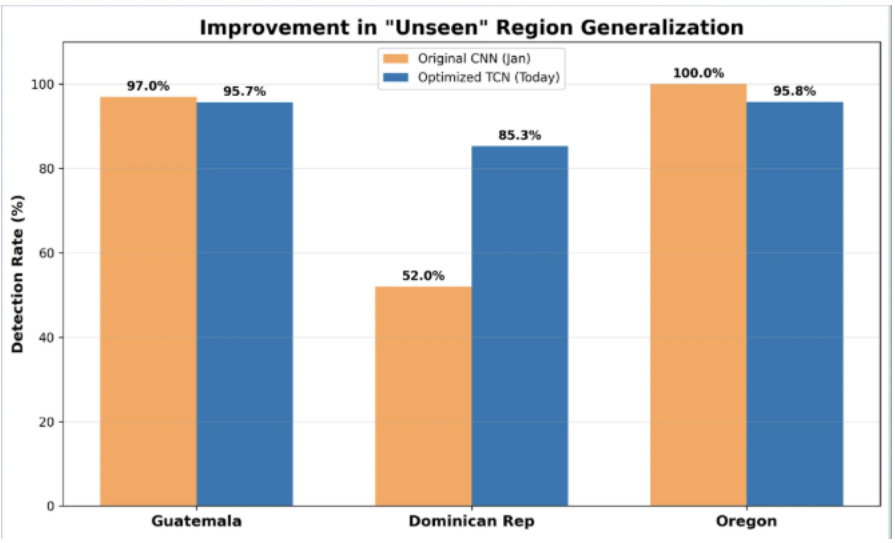

Figure 6: Multi-Regional Generalization: Original CNN vs. Optimized TCN.

What stands out most in these numbers is the false alarm rate. Oregon went from 10.2% down to 0.0%, and even the Dominican Republic, which has been the hardest region throughout this whole project, dropped from 7.6% to 1.5%. The 0.5 Hz high-pass filter was the key fix for Oregon specifically. Analysis of the previous failure cases showed that almost all of the false alarms there were caused by a slow baseline tilt in the MEMS sensor, exactly the kind of sub-0.5 Hz artifact that the filter removes. The 3-window temporal smoothing handled the rest by requiring three consecutive alarm windows before triggering, which filters out any single-spike mechanical noise.

The Dominican Republic achieving 85.3% detection is meaningful context too. The previous 3-location CNN baseline only managed 15% average confidence on Dominican Republic data. Expanding to 19 training locations brought that up to 52% confidence, and the full TCN pushed detection to 85.3%. That's a 5.9-fold improvement in average confidence from the baseline to the final model, a direct result of geographic diversity in the training data.

Casing Material Analysis

FEA results made the material choice straightforward. The PVC+ABS pairing showed Von Mises stress of 0.85 MPa compared to just 0.044 MPa for PVC+PET, that's 80.6% higher stress in the ABS version. Both are well below PVC's 55 MPa yield strength, but the gap in stress distribution efficiency is significant. Frequency analysis added more weight to the PET case: the PVC+PET system has a natural resonant frequency of 3,269 Hz, versus 7,493 Hz for the ABS version. The PET system's natural frequency is over 4,000 Hz lower, but it still sits thousands of times above the 0.1–10 Hz seismic range. Neither casing will resonate with but PET distributes the mechanical load more efficiently.

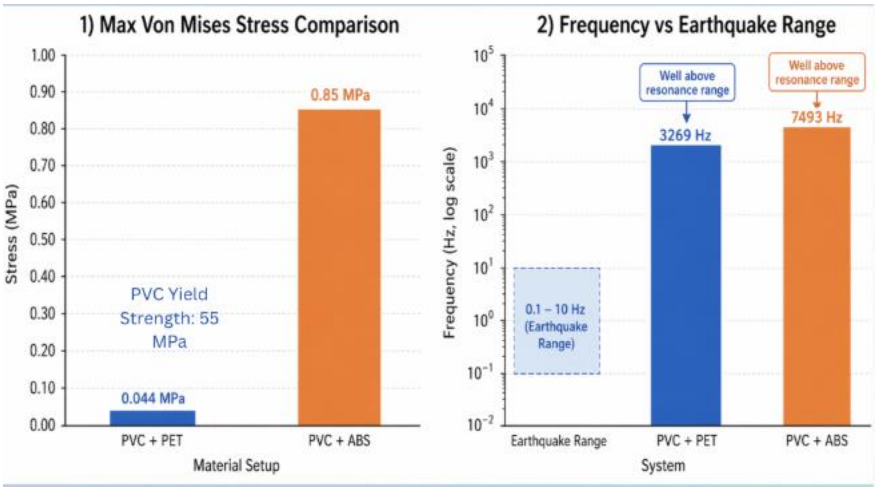

Figure 7: FEA Stress and Frequency Analysis — PVC+PET vs. PVC+ABS.

The stiffness argument clinches it. PET has a modulus of 2.7 GPa versus ABS's 1.8 GPa. A stiffer casing deforms less when the ground moves, which means it transmits ground motion to the accelerometer more faithfully. ABS would absorb a small but real portion of the vibration-essentially acting as a tiny shock absorber between the ground and the sensor. For a device whose entire job is to detect ground vibration, that's not acceptable. PVC+PET was the clear choice.

Conclusions

This project set out to answer a pretty direct question: can you build a seismic early warning system that actually works, for under $250, that could realistically be deployed in earthquake-prone developing communities? Based on our results, the answer is yes.

The final TCN model hit 100% detection in Oregon and Guatemala and 85.3% in the Dominican Republic, with false alarm rates near 1% in the primary test zones. The ADXL345 sensor validation showed that cheap hardware can measure high-impulse acceleration events with under 8% error, which puts it firmly in the range needed for seismic detection. The PVC+PET monocoque casing passed FEA with stress levels well under material limits and a natural frequency thousands of times above the seismic range. And the whole thing runs offline on a Raspberry Pi, with no cloud dependency, for 97% less than a commercial seismometer unit.

The biggest finding of this study is probably how much geographic diversity in the training data matters. The jump from a 3-location training set to 19 locations was responsible for most of the improvement in Dominican Republic generalization as it went from 15% to 52% average confidence before the architecture upgrade even happened. That result is consistent with what Kong et al. (2016) found about ML-based seismic detectors needing training data that matches the tectonic character of the target region.

There are real limitations to acknowledge. The Dominican Republic results, while much improved, still lag behind Oregon and Guatemala, probably because Caribbean plate seismicity is underrepresented in our training data relative to Pacific Rim events. The system also hasn't been validated on a shake table with controlled, known-amplitude inputs yet. That would let us properly

calibrate the threshold against standardized ground motion intensity measures. And integrated dual-sensor testing with the PZT is still pending due to the hardware constraints mentioned earlier.

Next steps are: shake table validation using a custom eccentric motor vibration platform, expansion of the training dataset with more Caribbean seismic events, PZT sensor soldering and noise injection testing, and eventual pilot field deployment to measure real-world performance. The foundation is solid. The path forward is clear.

References

Allen, R. M., & Melgar, D. (2019). Earthquake early warning: Advances, scientific challenges, and societal needs. Annual Review of Earth and Planetary Sciences, 47, 361–388. https://doi.org/10.1146/annurev-earth-053018-060457

Bose, M., Allen, R., Brown, H., Cua, G., Fischer, M., Hauksson, E., Heaton, T., Hellweg, M., Johnston, M., Maechling, P., Neuhauser, D., Pickles, P., Schwarz, S., Tinka, A., & Wurman, G. (2018). CISN ShakeAlert: An earthquake early warning demonstration system for California. Early Warning for Geological Disasters, 49, 1–22. https://doi.org/10.1007/978-3-662-45214-7_1

Center for Engineering Strong Motion Data. (2024). CESMD strong motion virtual data center. Retrieved from https://www.strongmotioncenter.org

Jaffe, H., & Berlincourt, D. A. (1965). Piezoelectric transducer materials. Proceedings of the IEEE, 53(10), 1372–1386. https://doi.org/10.1109/proc.1965.4253

Kohler, M. D., Smith, D. E., Andrews, J., Chung, A. I., Hartog, R., Henson, I., Lawrence, J. F., Bodin, P., & Meier, M. A. (2018). Earthquake early warning ShakeAlert 2.0: Public rollout. Seismological Research Letters, 89(6), 2490–2494. https://doi.org/10.1785/0220180245

Kong, Q., Allen, R. M., Schreier, L., & Kwon, Y. W. (2016). MyShake: A smartphone seismic network for earthquake early warning and beyond. Science Advances, 2(2), e1501055. https://doi.org/10.1126/sciadv.1501055

Lalithavani, K., Arasan, G., Premsundar, S., Venkatasubash, V., & Vijayaraj, R. (2025). AI-powered based real time earthquake early warning system using deep learning. Journal of

Engineering Research and Reports, 27(5), 433–442. https://doi.org/10.9734/jerr/2025/v27i51516

Mousavi, S. M., Ellsworth, W. L., Zhu, W., Chuang, L. Y., & Beroza, G. C. (2020). Earthquake transformer—An attentive deep-learning model for simultaneous earthquake detection and phase picking. Nature Communications, 11, 3952. https://doi.org/10.1038/s41467-020-17591-w

Perol, T., Gharbi, M., & Denolle, M. (2018). Convolutional neural network for earthquake detection and location. Science Advances, 4(2), e1700578. https://doi.org/10.1126/sciadv.1700578

United States Geological Survey. (2024). Earthquake hazards program. Retrieved from https://earthquake.usgs.gov

Wu, R., Selvadurai, P. A., Chen, C., & Moradian, O. (2021). Revisiting piezoelectric sensor calibration methods using elastodynamic body waves. Journal of Nondestructive Evaluation, 40(3). https://doi.org/10.1007/s10921-021-00799-1